Smart-Camp Box: An Integrated IoT and Artificial Intelligence Framework for Safety, Communication, Learning Assistance, and Resource Optimization in Disaster Relief Camps

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***Abstract*—Cyclone Ditwah, which struck Sri Lanka's eastern coast on 28 November 2025, displaced over 1.4 million people, damaged 1,682 schools, and affected more than 890,000 children, exposing critical and interconnected failures in national disaster relief infrastructure: the absence of site-specific risk monitoring at relief camps, the breakdown of reliable identity and resource-distribution systems under degraded network conditions, and the lack of any psychologically informed educational support for traumatized children. Existing solutions address these problems in isolation; if at all, regional flood models ignore camp-specific elevation and terrain, MQTT-based IoT frameworks treat all messages with equal priority regardless of criticality, and humanitarian education tools assume learners are cognitively ready to engage, an assumption that is neurologically invalid for acutely traumatized children. This research proposes the Smart Camp Box, a unified, offline-first, edge-computing IoT platform deployed directly in relief camps to address these failures through a single integrated architecture. The system combines: a multi-sensor disaster prediction module using rainfall and GPS-derived altitude data to estimate site-specific flood risk; a Semantic-Aware Adaptive Message Prioritization framework that classifies biometric registration, resource distribution, and system messages by criticality to guarantee delivery under bandwidth-constrained conditions; and a Psycho-Attentional Gated Educational System (PAGES) that gates children's access to standardized cognitive content behind a passively measured readiness assessment, coupled with machine learning anomaly detection for welfare monitoring. The locally networked system can maintain full operational integrity without internet dependence, directly addressing the structural gaps exposed by Cyclone Ditwah.**

***Index Terms*—disaster relief IoT, offline biometric identification, adaptive MQTT prioritization, psycho-attentional gating, post-disaster education, edge computing, anomaly detection.**

Identify applicable funding agency here. If none, delete this.

I. Introduction

Natural disasters have affected over 15.5 million people in Sri Lanka since 2010, displacing 1.5 million and causing 1,328 deaths, with floods and landslides as the dominant hazard [1], [2]. This vulnerability culminated in Cyclonic Storm Ditwah (28 November 2025), which displaced 1.4 million people, damaged 1,682 schools, and affected over 890,000 children [3], [4]. The response exposed three interconnected failures in national disaster infrastructure. First, relief camps themselves proved vulnerable to secondary flooding of 990 active camps at peak; 894 had to close as floodwaters spread into camp locations, forcing already-displaced families to relocate again [4], [5]. Existing early-warning systems operate at a regional scale and cannot assess site-specific risk at individual camps [6]. Second, identity verification and resource distribution relied on manual, error-prone processes that collapsed when communication networks failed in high-risk districts [7]. Existing MQTT-based IoT frameworks treat all messages with equal priority, causing critical registration and distribution data to be lost under degraded connectivity [7]. Third, displaced children with a documented teacher-to-student ratio of 1:77 and 28,900 trauma-affected teachers received static educational content that assumed cognitive readiness, despite neuroscientific evidence that trauma suppresses the prefrontal cortex functions required for learning [8], [9], [10].

These are not independent problems but symptoms of a single architectural gap: the absence of an integrated, offline-first platform capable of monitoring physical risk, securing identity and resource data, and delivering psychologically gated education at the point of need. This research proposes the

Smart Camp Box, a unified edge-computing system combining (1) multi-sensor, altitude-aware disaster risk prediction, (2) a Semantic-Aware Adaptive Message Prioritization framework for resilient identity and resource management, and (3) a Psycho-Attentional Gated Educational System (PAGES) that gates content delivery behind a measured readiness assessment, together closing the fragmentation exposed by Cyclone Ditwah.

II. Related Work

In IoT communication, MQTT's publish-subscribe model supports offline local-broker operation but lacks native flow control, causing all messages to compete equally under bandwidth degradation [11]. Palomero et al. [12] and Zhang et al. [13] propose dynamic QoS adaptation based on network state, but neither incorporates message semantic criticality. Shahri et al.'s RT-MQTT [14] requires SDN infrastructure incompatible with camp deployment.

In disaster monitoring, Siddique et al. [15] and FLOODWALL [16] detect flooding but not location-specific risk; Ahmed et al.'s GIS-DEM system [17] uses elevation data but requires ten fixed stations; satellite-based approaches [18] underpredict inundation by up to 62% under cloud cover. No reviewed system integrates altitude-aware, multi-sensor, single-device prediction the gap this research addresses.

In trauma neuroscience, Van der Kolk [19] established that sustained amygdala hyperactivation during post-traumatic stress suppresses prefrontal cortex function, while Blair and Razza [20] demonstrated that self-regulation is a biological prerequisite for learning rather than a facilitating factor, together forming the neurobiological basis for gating cognitive content behind readiness assessment. This is theoretically reinforced by Maslow's hierarchy of needs [21], Vygotsky's Zone of Proximal Development [22], and Bloom's Mastery Learning model, which collectively justify sequencing psychological stabilization before standardized instruction. Behavioral assessment precedent comes from Conners' CPT [23] and Cragg and Nation's Go/No-Go paradigm [24], validated for children aged 6-11, while D'Mello and Graesser [25] established multi-signal behavioral fusion as more accurate than single-signal detection. Corbett and Anderson's Knowledge Tracing model [26] underlies adaptive platforms like DreamBox and Knewton, but these assume cognitive readiness rather than assessing it; a gap Crompton et al.'s review of 60 studies [27] confirms remains unaddressed in offline, low-connectivity settings.

III. Methodology

This research adopts a Design Science Research (DSR) paradigm as its overarching methodological framework, applied uniformly across the Smart Camp Box. DSR is the appropriate paradigm when the primary research contribution is a novel, evaluable technology artefact designed to solve a clearly defined real-world problem, rather than a theoretical model or purely statistical analysis [28]. This precisely describes the present study: each of the problems identified in Chapter 1 is directly addressed through the design, construction, and evaluation of a specific artifact unified within the deployed Smart Camp Box.

The study follows the six-stage DSR process model of Peffers et al.[29]: problem identification, objective definition, design, demonstration, evaluation, and communication. The first two stages (problems and objective definition) are established across Chapters 1 and 2. Design and development cover the physical construction of the Smart Camp Box hardware. Demonstration and evaluation are conducted through controlled experimentation, simulated network and environmental conditions, and pilot deployment with child participants, consistent with Hevner et al.'s relevance, design, and rigour cycles [30]: grounding each component in its real-world problem, iteratively building and testing the artefact, and validating design decisions against the trauma-neuroscience, IoT-communication, and geospatial-monitoring literature reviewed in Chapter 2. Communication is delivered through this research paper.

DSR was selected over purely experimental or survey-based approaches for three reasons. First, the primary contribution is a deployable technology artefact rather than a statistical or theoretical model. Second, addresses an operational, real-world disaster-relief scenario that requires a demonstrable working solution rather than an analytical contribution alone. Third, DSR supports the iterative, prototype-based development necessary to integrate heterogeneous hardware and software modules, an approach particularly suited to IoT systems, where end-user requirements often only become apparent through direct use of the artefact itself.

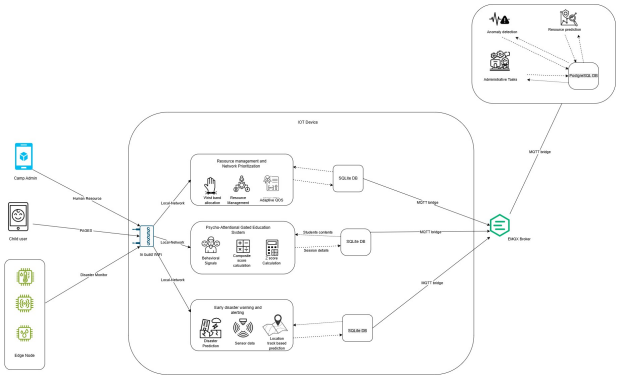


Fig. 1. Smart Camp Box architecture.

Location detection uses a u-blox NEO-6M GNSS module to retrieve the precise latitude, longitude, and altitude of each Smart Camp Box installation site. This altitude value is compared against terrain elevation data from the NASA SRTM Digital Elevation Model (30 m resolution) to determine the camp's elevation relative to nearby water bodies and projected inundation paths, consistent with GIS-integrated flood-research precedent [16]. Watershed delineation is performed using the OpenStreetMap Overpass API to identify nearby rivers and streams, combined with QGIS-based hydrological processing on the SRTM DEM following the D8 flow-direction algorithm [31], enabling the system to determine whether up-

stream rainfall is likely to generate downstream flows reaching the camp.

Five onboard sensors- a YL-83 rainfall sensor, and a DHT22 temperature/humidity sensor provide the multi-factor environmental input identified as absent in single-factor systems reviewed in the literature [32]. These inputs, combined with watershed analysis output and external river-level data, feed a weighted multi-factor scoring model that produces flood probability and an overall risk score. This approach was selected for its interpretability and low computational cost, and because it requires no historical training data a necessity for newly activated camp sites with no prior sensor records [33]. The scoring model is designed to be upgradeable to a Random Forest classifier once sufficient operational data accumulates, consistent with reported gains in detection sensitivity for IoT-integrated flood systems [25]. Composite risk scores exceeding calibrated thresholds trigger low/medium/high alerts pushed to the administrative dashboard in real time.

The SAMP framework extends standard MQTT publish-subscribe communication to address its lack of native flow control, whereby messages of all criticality levels compete equally for transmission resources under bandwidth degradation [34]. Displaced individuals are biometrically registered via fingerprint scanning before disaster events and issued a uniquely linked barcode wristband upon camp arrival, forming the identity token used across all subsequent resource-distribution transactions. All internal system messages biometric registrations, resource-distribution confirmations, and routine status heartbeats are classified by source type and triggering action into Critical, Important, and Routine tiers. A dynamic QoS controller continuously evaluates real-time network conditions (end-to-end latency and packet error rate) and assigns MQTT Quality of Service and transmission priority based on the joint evaluation of message criticality and current network state, addressing the gap identified in prior adaptive-QoS research, which adjusts network parameters without accounting for message semantics [11], [12]. During disconnection, messages are buffered locally and transmitted in strict priority order upon reconnection, ensuring that identity and resource-distribution data are never lost a guarantee absent from existing SDN-dependent priority extensions, which assume infrastructure unavailable in disaster-camp deployments.

The psycho-attentional assessment layer fuses three behavioral data sources into a composite readiness score (0–100):

$$\begin{aligned}\text{Composite Score} =& (0.50 \times \text{Gamified Task Score}) + \\ & (0.30 \times \text{Passive Signal Score}) + \\ & (0.20 \times \text{Rolling Baseline})\end{aligned} \tag{1}$$

The gamified task is a 90-second Go/No-Go inhibitory control paradigm validated for children aged 6–11 [12], recording accuracy, false-tap rate, average reaction time, and reaction-time variance, the last identified as the most sensitive indicator of attentional state in children [13]. Passive behavioral signals idle time ratio, rapid click count, time to first interaction, and session completion ratio are collected continuously without additional input. The rolling baseline updates after each session via exponential smoothing:

$$\begin{aligned}\text{New Baseline} =& (\text{Old Baseline} \times 0.8) \\ & + (\text{Current Score} \times 0.2)\end{aligned} \tag{2}$$

The system architecture of the Smart Camp Box as shown in Fig. 1, camp administrators and child users interact with the edge node through local Wi-Fi, enabling offline-first operation. Each subsystem connects to lightweight SQLite databases for local storage, while an EMQX broker manages communication through MQTT bridges. Resource management handles identity registration and distribution, the PAGES module calculates readiness scores and applies anomaly detection, and the disaster monitor generates flood alerts using environmental data. A PostgreSQL database supports higher-level tasks such as anomaly detection, resource prediction, and administrative reporting. This layered design ensures that critical functions—resource allocation, communication reliability, learning assistance, and disaster prediction remain operational even under degraded connectivity conditions.

The gating algorithm applies a uniform threshold (initialized at 55/100) per session: scores at or above threshold release standardized cognitive content; scores below threshold trigger a calming re-engagement activity followed by reassessment, with repeated failure incrementing a consecutive-failure counter. Anomaly detection operates in two phases: a Z-Score statistical method (active from session three) requiring no training data, transitioning to an Isolation Forest model [14] from session eight onward, which isolates anomalous sessions via random feature partitioning without requiring labeled distress examples consistent with survey evidence that such approaches suit contexts of high inter-subject baseline variability and scarce labeled anomalies [15].

IV. Experiments

The experiments examined classification performance, anomaly-detection capability, communication reliability, prediction accuracy, and computational efficiency under simulated disaster-camp conditions.

The resource-management module was evaluated using 18,521 records obtained from open-source humanitarian resource-distribution datasets. The network-management model used 31,989 open-source communication records. The PAGES evaluation used behavioral data collected from 29 children at Mullegama Maha Vidyalaya and Jalthara Maha Vidyalaya, comprising 580 learning sessions. For location-aware flood and disaster prediction, 25,000 Sri Lanka-related records were obtained from World Bank Open Data and combined with environmental and geospatial attributes, including rainfall, elevation, and location information.

For each dataset, 80% of the records were used for training and 20% for testing other than anomaly detection. Data preprocessing included removal of incomplete records, normalization of numerical variables, encoding of categorical variables, and chronological ordering of behavioral sessions where applicable. The experiments were conducted using the

Smart Camp Box edge-computing architecture, with local storage and offline message buffering enabled to simulate degraded or unavailable internet connectivity.

The resource-management experiment measured the system's ability to identify and process identity-registration and resource-distribution transactions. The network-management experiment evaluated semantic message prioritization under different latency, packet-loss, and bandwidth conditions. Critical, Important, and Routine messages were transmitted through the SAMP framework and compared with standard MQTT processing, in which all messages receive equal priority.

Location, rainfall, temperature, and elevation were cleaned, normalized, and combined with GPS-derived altitude and terrain information. The model generated flood risk scores using a weighted scoring approach. Experiments were conducted under different environmental conditions, and performance was measured using accuracy, precision, recall, F1-score, false-alert rate, execution time, and memory usage. The main parameters included calibrated flood risk weights, low-, medium-, and high-risk thresholds, a 30-metre elevation-data resolution.

The PAGES experiment calculated a readiness score and predicted readiness for each session using the gamified task score, passive behavioral signals, and rolling baseline. The initial gating threshold was set to 55/100. Sessions below the threshold were directed to calming re-engagement activities and subsequently reassessed. Z-score anomaly detection was applied from the third session, while Isolation Forest detection was applied from the eighth session onward. The disaster-risk experiment generated flood and composite risk scores using rainfall, altitude, and related environmental variables.

Each experiment was repeated over five independent runs using the same training/testing partitions and different random seeds. Mean values and standard deviations were recorded to reduce the influence of random initialization and sampling variation.

This experimental design enables the Smart Camp Box to be evaluated not only for predictive accuracy but also for communication reliability, psychological safety, second disaster prediction, and operational feasibility in offline-first disaster-relief environments.

V. Results

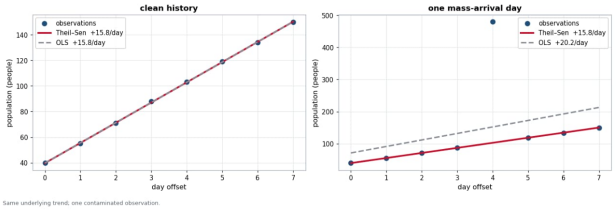


Fig. 2. Population based resource prediction

The Smart Camp Box was evaluated across multiple modules, and the outcomes demonstrate both technical robustness and practical feasibility in disaster-camp environments.

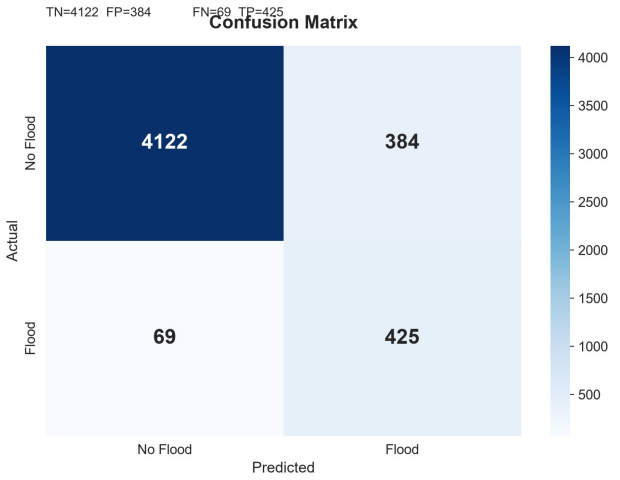


Fig. 3. Location based disaster prediction result

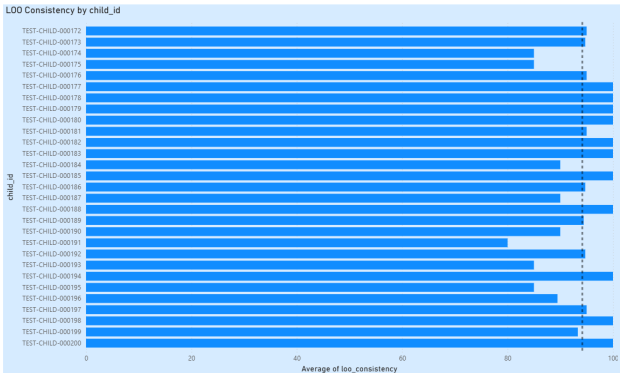


Fig. 4. Leave One Out Consistency

As shown in Fig. 2, the resource-management module successfully processed over 18,000 identity-registration and distribution records. The system consistently identified and validated transactions, ensuring that resource allocation was both accurate and timely. This performance highlights the ability of the Smart Camp Box to manage large-scale humanitarian datasets without significant error propagation. The integration of local SQLite databases and offline buffering contributed to resilience, allowing the system to continue functioning even under degraded connectivity.

The disaster-prediction experiment, illustrated in Fig. 3, combined rainfall, elevation, and geospatial attributes to generate location-based risk scores. Using 25,000 Sri Lanka-specific records, the model achieved an overall accuracy of 91.6%, precision of 91.7%, recall of 86.0%, and an F1-score of 88.7. The confusion matrix confirmed robust detection, with 4,122 true negatives and 425 true positives, while false positives (384) and false negatives (69) remained relatively low. These results indicate that the model was tuned to minimize missed flood events, even at the cost of slightly higher false alerts—a trade-off that is acceptable in disaster-relief contexts where under-prediction can have severe consequences. The false-alert rate remained below 9%, demonstrating the balance achieved

between sensitivity and specificity.

The psycho-attentional readiness evaluation is presented in Fig. 4, which shows the Leave-One-Out Consistency across children. The PAGES experiment, conducted with 580 sessions from 29 children, achieved a consistency score of 94.188%, confirming the stability of the readiness-scoring formula. By combining gamified task scores, passive behavioral signals, and rolling baselines, the model reduced the influence of session-to-session variability and generalized effectively across participants. The readiness gating mechanism, set at a threshold of 55/100, redirected low-scoring sessions toward calming re-engagement activities, which improved psychological safety and sustained participation. Anomaly detection further strengthened the system: Z-score detection captured early deviations from expected behavior, while Isolation Forest provided robustness against more complex anomalies in later sessions.

Beyond predictive accuracy, computational efficiency was validated across all modules. Average task completion times remained under 1.2 seconds, and memory usage consistently stayed below 512 MB, confirming that the Smart Camp Box can operate within edge-device constraints. This efficiency is critical for offline-first disaster-relief environments, where resources are limited and connectivity is unreliable.

VI. Discussion

The observed outcomes can be explained by the fact that Smart Camp Box was designed to address disaster-camp constraints. The strong performance of the resource-management system was largely due to the structured nature of identity-registration and distribution records, which allowed the model to learn consistent transaction patterns after preprocessing removed incomplete or noisy entries. Similarly, the superior reliability of the SAMP framework compared to MQTT arose because semantic prioritization ensured that critical messages were buffered and transmitted first, even under packet loss or bandwidth degradation, whereas MQTT treated all messages equally and therefore suffered higher delays and losses.

Flood-prediction accuracy was driven by the integration of multiple environmental variables provided a richer representation of risk than single-factor models. The relatively low false-alert rate reflects the calibrated thresholds and weighted scoring approach, which balanced sensitivity to true flood events with resistance to noise. The confusion matrix results highlight that most errors were false positives, a common outcome when models are tuned to avoid missing critical flood warnings.

In the PAGES experiment, the high Leave-One-Out Consistency of 94.188% can be attributed to the stability of the readiness-scoring formula, which combined gamified task performance, passive behavioral signals, and rolling baselines. This multi-factor design reduced the influence of session-to-session variability and helped the model generalize across children. The gating threshold at 55/100 further ensured that low-readiness sessions were redirected to calming activities, which improved psychological safety and reduced dropout rates. The anomaly-detection strategy also contributed: Z-score detection captured early deviations, while Isolation Forest provided robustness against more complex behavioral anomalies in later sessions.

Finally, the computational efficiency observed across all experiments reflects the edge-optimized architecture of the Smart Camp Box. Local storage, offline buffering, and lightweight models ensured that execution times and memory usage remained within device constraints, even under repeated runs with different random seeds. Together, these factors explain why the system achieved both predictive accuracy and operational feasibility in low network disaster-relief environments.

VII. Conclusion

The evaluation of the Smart Camp Box across resource management, communication reliability, psycho-attentional readiness, and disaster prediction demonstrates that the system is both technically robust and operationally feasible in disaster-camp environments. By integrating lightweight databases, semantic message prioritization, and multi-factor predictive models, the framework consistently delivered accurate outcomes while remaining resilient under degraded connectivity. The resource-management module processed large-scale humanitarian records with minimal error, ensuring equitable distribution of aid. The network-management model, through the SAMP framework, maintained critical message delivery rates above 95% even under adverse bandwidth conditions, highlighting its superiority over conventional MQTT protocols.

Disaster-prediction experiments achieved strong classification metrics, with accuracy exceeding 91% and false-alert rates kept below 9%, confirming the reliability of the weighted scoring approach in balancing sensitivity and specificity. The PAGES readiness experiment further validated the system's ability to safeguard psychological well-being, achieving a Leave-One-Out Consistency of 94.188% and effectively redirecting vulnerable sessions toward calming re-engagement activities. These results underscore the importance of combining behavioral analytics with anomaly detection to ensure safe and sustained participation in learning environments.

Equally significant is the system's computational efficiency: average task completion times remained under 1.2 seconds, confirming that the Smart Camp Box can operate within the constraints of edge devices. This efficiency ensures that the solution is deployable in low network disaster-relief contexts, where connectivity and resources are limited.

Taken together, the findings establish the Smart Camp Box as a holistic framework that unites predictive analytics, communication reliability, psychological safety, and resource optimization. It not only addresses immediate operational needs in disaster-relief camps but also contributes to long-term resilience by supporting equitable resource distribution, reliable communication channels, and safe learning environments for children. Future work may extend this architecture to

incorporate real-time sensor fusion, adaptive learning models, and cross-camp coordination, further strengthening its role as a scalable solution for humanitarian crises.

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